PROJECT CONCEPT NOTE

Exmentify

AI-Powered Exam Prep Companion

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Live Application

Primary URL (HTTPS via CloudFront):
https://d1vcdm3hqyivum.cloudfront.net/

Backend URL (HTTP via AWS Elastic Beanstalk):
http://sheema-exmentify-app.ap-south-1.elasticbeanstalk.com/

Note: The application is hosted on AWS Elastic Beanstalk. A CloudFront distribution sits in front of it to serve the app over HTTPS. Both URLs are live and point to the same application.

To see live AI-generated responses, click the GATE - Computer Science (CS) card on the homepage. The other exam cards (other GATE branches, UGC NET, CAT, GRE, etc.) are shown as "Coming Soon" to demonstrate that the application's architecture is exam-agnostic, but only GATE CS has topics, PYQs, and AI generation fully wired up in this submission.

1. Project Title & Application Name

Project Title	Exmentify — AI-Powered Exam Prep Companion
Application Name	Exmentify
Built For	IBM SkillsBuild × BharatCares (AICTE) — Final Project Submission

2. Problem Statement / Objective

Competitive-exam aspirants in India (GATE, and similarly structured exams) rely on scattered resources — static PDFs, generic YouTube explainers, and question banks with no personalized feedback loop. Revision is inefficient because students cannot quickly get a concept re-explained at their own level, in their own language, or diagnose exactly why a specific answer was wrong.

Exmentify's objective is to give every aspirant a single, on-demand AI study companion that explains concepts, generates revision notes, creates practice questions, solves previous-year questions, and

diagnoses individual mistakes — all in real time, in the language and difficulty level the student chooses, without requiring a human tutor or a paid coaching subscription.

3. Target User & Use Case

- **Primary user:** Undergraduate engineering students preparing for GATE Computer Science, typically in their final year, self-studying alongside coursework.
- **Secondary users:** Aspirants for other competitive exams — the architecture is exam-agnostic, with additional exams wired in as "Coming Soon" to demonstrate the design generalizes beyond GATE CS.
- **Core use case:** A student picks an exam and topic (e.g., GATE CS → Data Structures & Algorithms → Trees), then chooses a mode — Explain, Revision Notes, Practice, PYQ Solutions, Ask a Doubt, Flashcards, or Diagnose Mistake — and receives a streamed, personalized AI response instead of searching through static material.
- **Secondary use case:** Weekly low-pressure contest mode for self-testing, and a study-plan generator that lays out a day-by-day schedule against the student's actual exam date.

4. LLM Model & API Used

Provider	Google AI (Gemini API)
Model	gemini-flash-lite-latest
Integration	Server-Sent Events (SSE) streaming from a FastAPI backend, so responses render token-by-token in the UI rather than as a single blocking response
Why this model	Low latency and cost-efficient for a high-frequency, conversational study-assistant workload, while remaining capable enough for structured explanations, code-adjacent DSA content, and multi-language (English / Hindi / Hinglish) output

5. Key Features of the Application

- **Explain:** Concept explanations in English, Hindi, or Hinglish, streamed live — with an inline Mermaid diagram automatically included when a visual genuinely aids understanding (e.g. trees, graphs, state machines).
- **Revision Notes:** Condensed, exam-focused notes generated per topic.
- **Practice:** AI-generated practice questions at Easy / Medium / Hard difficulty.
- **PYQ Solutions:** Step-by-step solutions to previous-year questions, picked from a per-topic question bank.

- **Ask a Doubt:** A focused follow-up chat scoped to the current topic.
- **Flashcards:** Auto-generated flip-through flashcard sets per topic.
- **Diagnose Mistake:** Student submits a question, their answer, and (optionally) the correct answer; the AI explains exactly where the reasoning went wrong.
- **Weekly Contest Mode:** Timed, topic-selectable practice test with personalized feedback — deliberately non-comparative (no leaderboard or rank), to keep the focus on self-improvement.
- **Structured Cheat Sheets:** Quick-reference cards per subject, organized by topic with labeled high-yield facts.
- **Study Plan Generator:** Builds a day-by-day revision schedule against the student's actual exam date, downloadable as a PDF.
- **Progress & Motivation:** Daily goal ring, study streak tracking, badges, bookmarking, and a lightweight name-entry personalization — all without requiring account creation.
- **PDF Export:** Notes, cheat sheets, and study plans can all be downloaded as PDFs for offline revision.

6. Expected User Experience & Outcomes

The experience is designed to feel like a fast, always-available personal tutor rather than a static content library. A student should be able to land on a topic, get an explanation in the language and depth they need within seconds (via live token streaming rather than a loading spinner), and immediately follow up with a doubt, a practice set, or a mistake diagnosis — all inside one continuous workspace, with zero login friction.

Expected outcomes for the student:

- Faster revision cycles, since explanations and notes are generated on demand instead of searched for.
- Better retention of weak areas through the Diagnose Mistake and Practice modes, which target the specific gap rather than generic drilling.
- Reduced exam anxiety, since contest mode gives feedback without a public leaderboard or comparative ranking.
- A visible sense of progress via streaks, badges, and a daily goal ring, encouraging consistent daily study rather than last-minute cramming.

At a technical level, the deliverable demonstrates a full-stack, production-deployed AI application: a FastAPI backend orchestrating Gemini API calls over SSE, a vanilla-JS frontend consuming that stream in real time, and a Dockerized deployment reachable over a public HTTPS URL — fulfilling both the functional and deployment expectations of the IBM SkillsBuild × BharatCares (AICTE) brief.